



the caffeine crew

Group 1A

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Meet Our Team



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2. Brief Description of data, model & analyses
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Outline of the Problem / Marketing Question

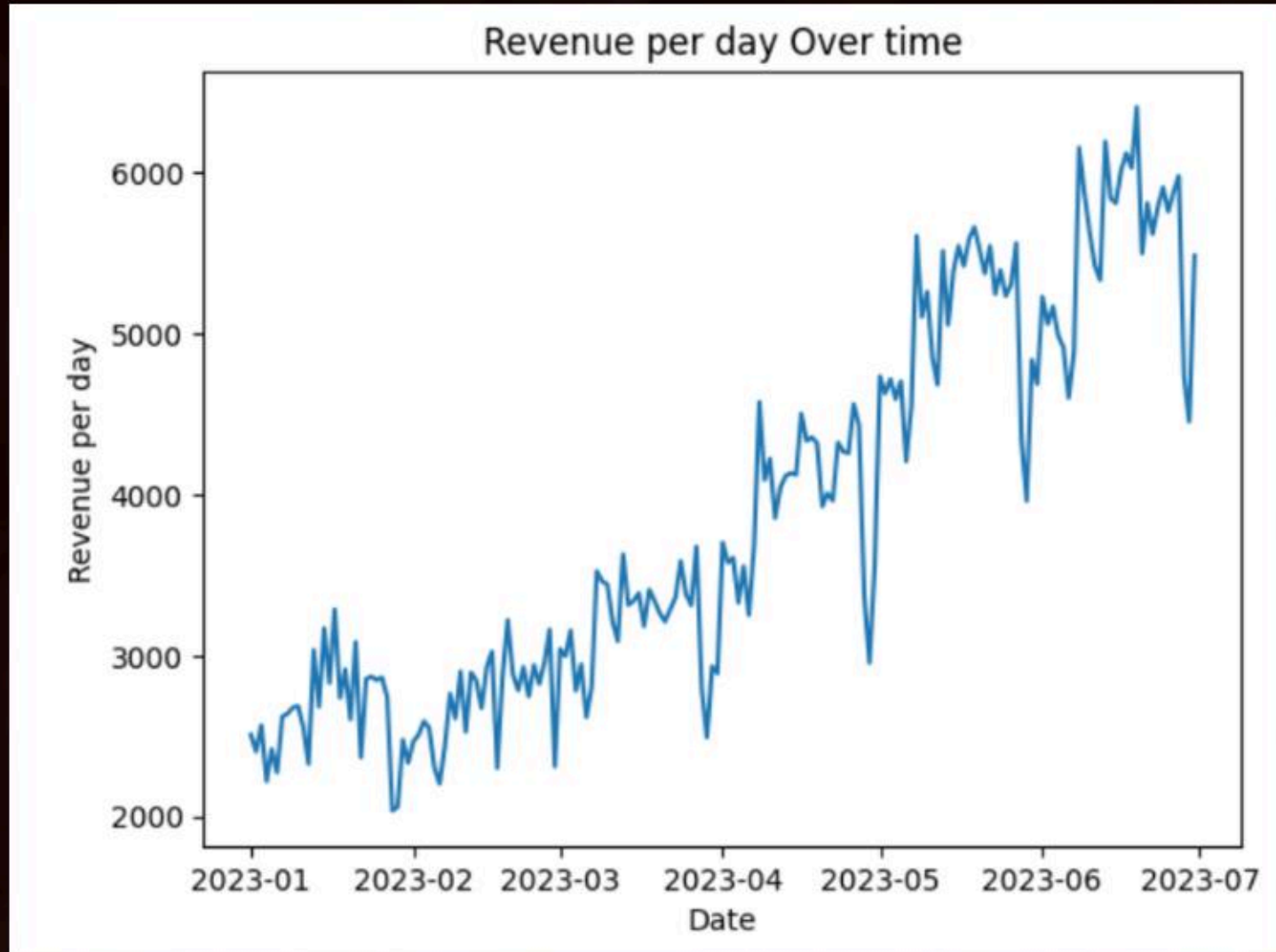
How can a coffee shop optimize the timing and type of promotions to increase customer engagement and sales ?



Our Dataset



- **Dataset source:** Comprehensive transaction data from Kaggle, covering three coffee shop locations in New York City
- **Time coverage:** Sales data exclusively from 2023, including transaction date, time, product types, and store locations
- **Dataset scale:** Over 149,000 transactions providing rich insights into customer purchasing behavior
- **Key variables:** Transaction timing (hour of day, day of week), product categories, sales volume, and location-specific patterns
- **Project objective:** Identify optimal promotional timing and design targeted offers aligned with customer preferences and peak sales periods



- **Overall trend:** Daily revenue shows consistent upward growth throughout 2023
- **Starting point:** Revenue began at approximately \$2,000-\$3,000 per day in early 2023
- **Ending point:** Revenue climbed to \$5,000-\$6,000 per day by year-end

Revenue Over Time

01 Key insight:

Data table reveals specific dates with significant revenue changes—both peaks and drops

02 Strategic value:

Pattern indicates growing customer demand and identifies optimal periods for promotions and inventory planning

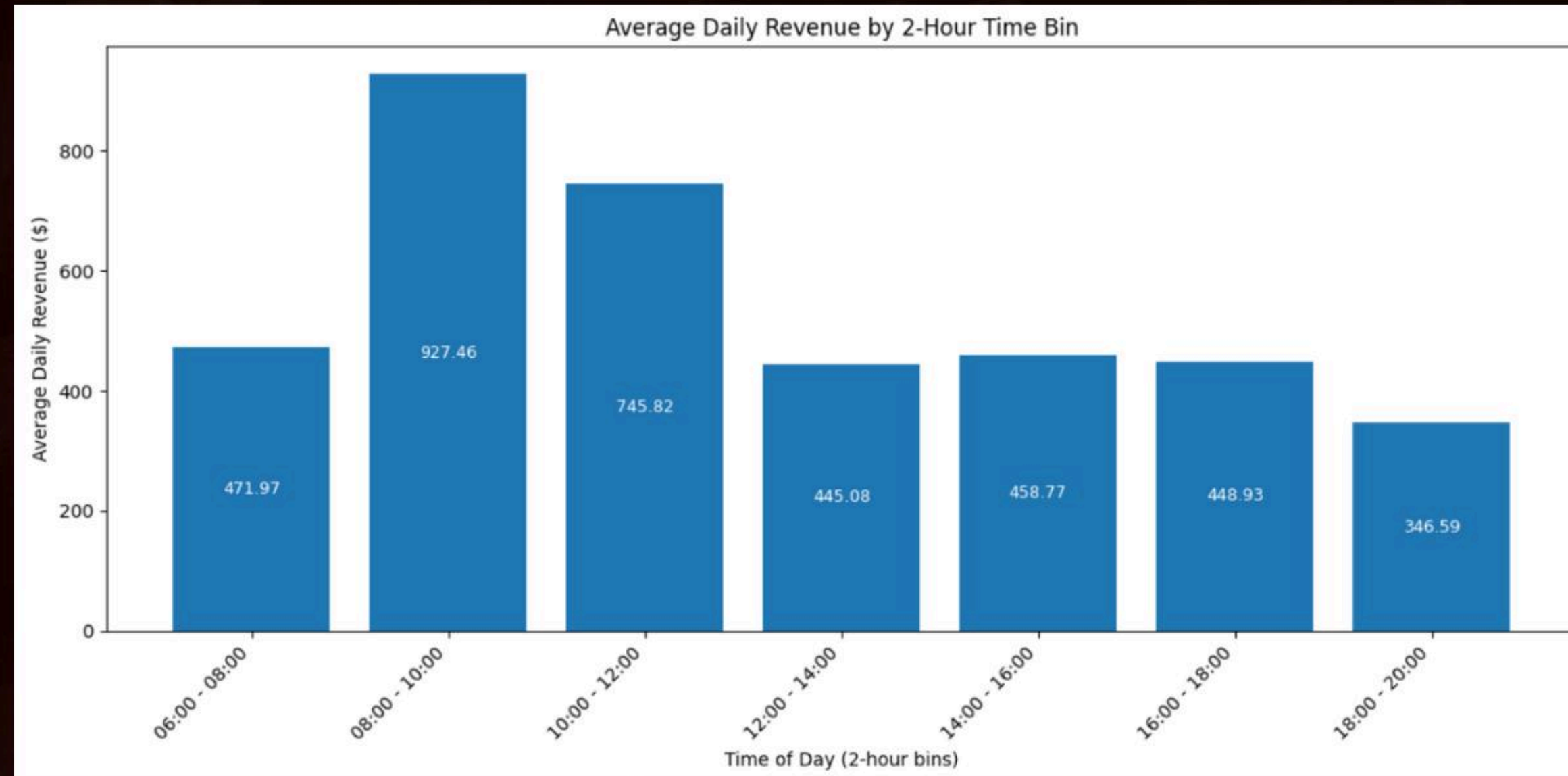
Revenue by Store Location



- **Hell's Kitchen leads:** Highest revenue at \$236,511, showing strongest market performance
- **Close competition:** Lower Manhattan (\$236,057) and Astoria (\$232,243) show similar performance levels
- **Consistent demand:** Relatively balanced revenue across all three locations indicates city-wide customer base
- **Strategic insight:** Hell's Kitchen and Lower Manhattan could benefit from additional resources or targeted promotions

Peak Hours Analysis

- **Morning rush dominates:** 8 AM-10PM generates highest revenue (\$927), indicating strong breakfast/brunch demand
- **Afternoon steady performance:** 12-6 pm shows consistent revenue around(\$450)
- **Evening sharp decline:** Revenue drops significantly after 6pm showing lowest performance (~\$346.59))



Strategic opportunity:

Optimize staffing and promotions for morning peak hours; test afternoon/evening initiatives to boost slower periods

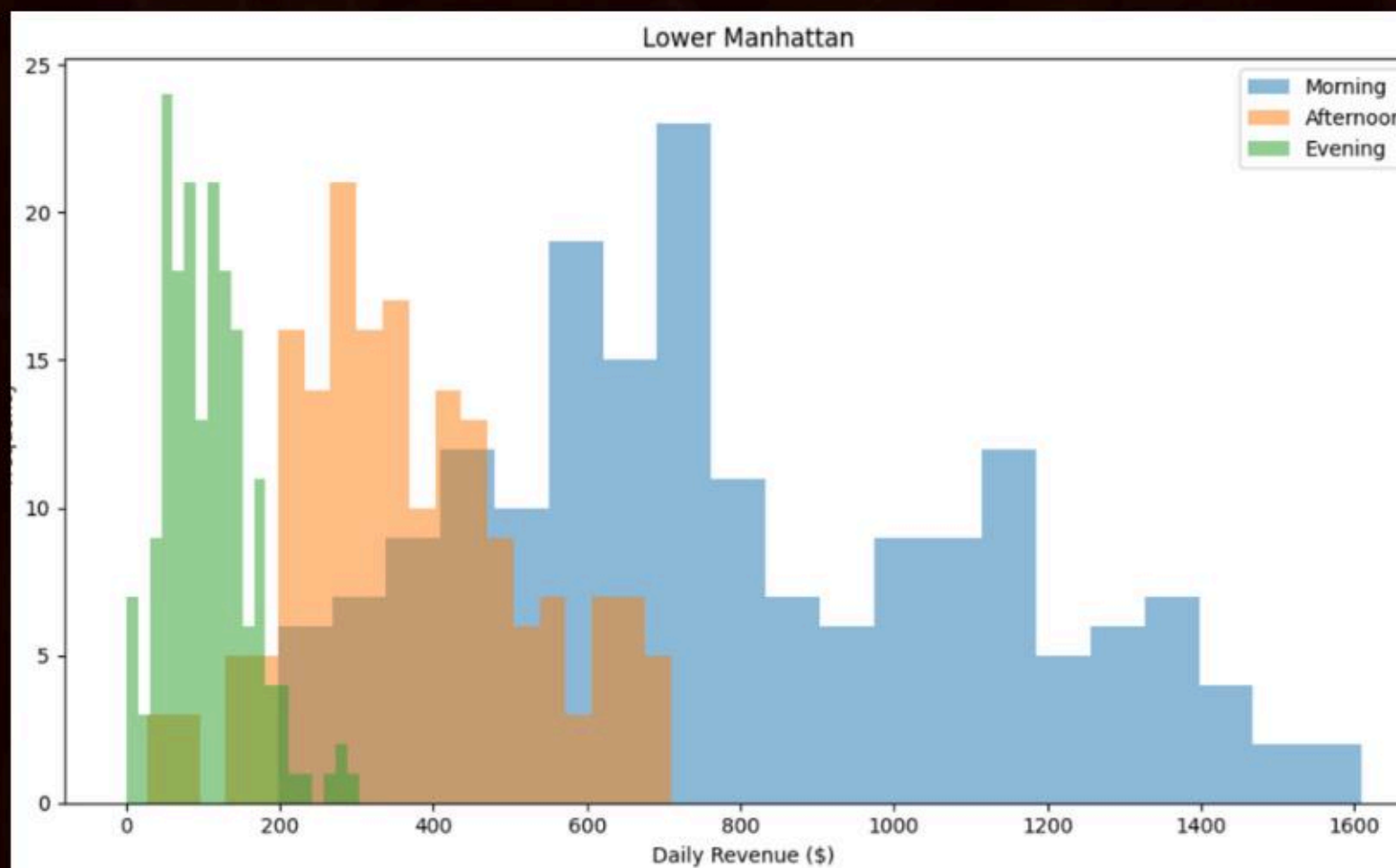
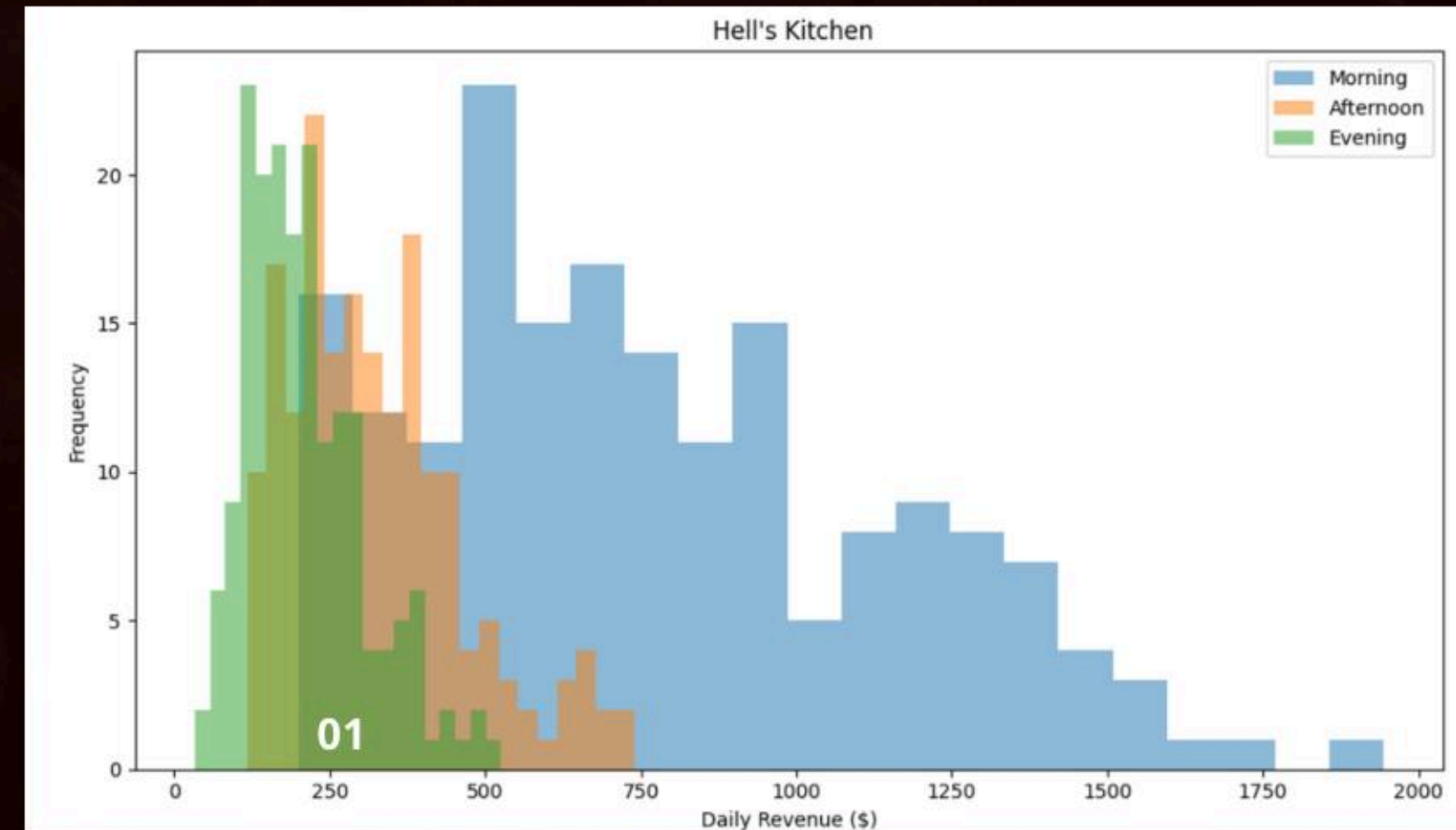
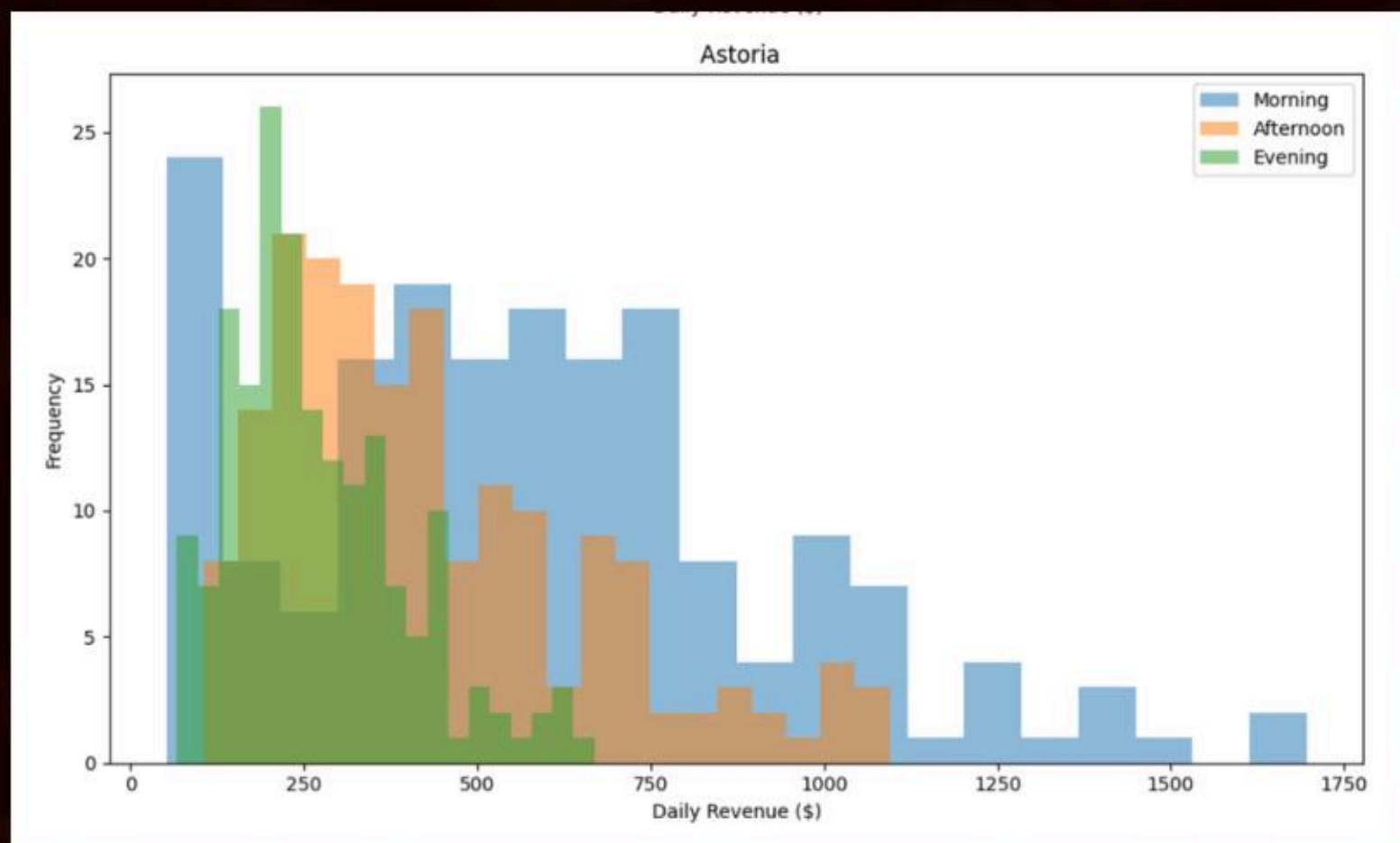
Product Category Heat Map



Morning is the strongest sales period overall, especially for Coffee and Tea, which show the highest volumes. Most categories peak during this time, indicating heavy morning customer demand.

Sales decline steadily through the afternoon and drop further in the evening across nearly all product categories. This suggests that morning and early afternoon drive the majority of café revenue, while evenings are low-traffic.

Revenue Layered Histogram



These graphs show the distribution of daily revenue in all 3 stores, by frequency, at three different times of the day.

Analysis:

Hell's kitchen underperforms in the afternoon as compared to the other stores.

Astoria performs better in the morning and afternoon as compared to the other stores.

Strategic Insight:

See what Astoria is doing differently in the afternoons and evenings as compared to the other stores, and try to reconcile.

These graphs can also help managers staff at the three stores based on individual store demand/revenue.

01

Revenue shows statistically significant growth throughout 2023 ($p < 0.001$)

02

Strongest sales day, while Tuesday shows lowest revenue among weekdays

03

Monthly analysis reveals consistent growth with peaks in specific months

04

Weekly patterns enable targeted promotions on slower days (e.g., Tuesday) and optimized staffing for high-volume days (e.g., Monday)

OLS Regression Results						
=====						
Dep. Variable:	Revenue	R-squared:	0.900			
Model:	OLS	Adj. R-squared:	0.893			
Method:	Least Squares	F-statistic:	125.6			
Date:	Wed, 19 Nov 2025	Prob (F-statistic):	3.07e-77			
Time:	22:44:40	Log-Likelihood:	-1329.6			
No. Observations:	181	AIC:	2685.			
Df Residuals:	168	BIC:	2727.			
Df Model:	12					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	2462.4197	111.954	21.995	0.000	2241.402	2683.437
t	9.0087	3.324	2.711	0.007	2.447	15.570
DayOfWeek_Friday	31.2965	108.122	0.289	0.773	-182.156	244.749
DayOfWeek_Monday	88.3518	108.047	0.818	0.415	-124.953	301.657
DayOfWeek_Saturday	50.4327	109.088	0.462	0.644	-164.928	265.793
DayOfWeek_Thursday	17.0276	108.134	0.157	0.875	-196.448	230.503
DayOfWeek_Tuesday	-6.0931	108.081	-0.056	0.955	-219.464	207.278
DayOfWeek_Wednesday	17.0182	108.123	0.157	0.875	-196.436	230.473
Month_2	-181.1369	141.221	-1.283	0.201	-459.934	97.660
Month_3	22.4795	219.825	0.102	0.919	-411.496	456.455
Month_4	523.7750	313.870	1.669	0.097	-95.863	1143.413
Month_5	1339.3703	410.968	3.259	0.001	528.044	2150.696
Month_6	1559.1387	510.171	3.056	0.003	551.967	2566.310
=====						
Omnibus:	18.219	Durbin-Watson:	1.015			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	20.760			
Skew:	-0.743	Prob(JB):	3.10e-05			
Kurtosis:	3.740	Cond. No.	2.74e+03			
=====						

Results

- **Revenue is rising throughout the year**

Our time-series analysis shows a clear upward trend in 2023, indicating stable and growing customer demand.

- **Weekly sales patterns are strong and consistent**

Mondays have the highest revenue, while Tuesdays perform the weakest. These patterns repeat across multiple months.

- **Seasonal peaks exist**

Certain months (e.g., May–June) show significantly higher revenue, confirming seasonal demand cycles.

- **Insights support actionable decisions**

The identified trends can guide promotion timing, staffing adjustments, and store-level targeting to increase overall revenue.



Product Category Regression

01 Packaged chocolate impact:

Each additional unit sold correlates with \$30.66 less revenue (cannibalization effect)

02 Coffee beans drive revenue:

Strongest positive correlation with higher revenue among all product categories

03 Time-of-day patterns:

Evening sales generate more revenue; morning sales generate second-most; afternoon

04 generates least

High model accuracy:

Predictor variables explain 98.3% of daily revenue variation ($R^2 = 0.983$)

OLS Regression Results

```
=====
Dep. Variable:      Revenue      R-squared:      0.983
Model:              OLS          Adj. R-squared:  0.982
Method:             Least Squares  F-statistic:    966.7
Date:               Wed, 19 Nov 2025  Prob (F-statistic): 3.56e-144
Time:               22:26:32      Log-Likelihood: -1170.5
No. Observations:   181          AIC:             2363.
Df Residuals:       170          BIC:             2398.
Df Model:           10
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	13.7108	42.173	0.325	0.745	-69.539	96.960
TimeOfDay_Evening	2.5901	1.004	2.579	0.011	0.608	4.572
TimeOfDay_Morning	2.0246	0.481	4.206	0.000	1.074	2.975
product_category_Brande	16.8161	4.391	3.829	0.000	8.148	25.485
product_category_Coffee	4.5608	0.744	6.130	0.000	3.092	6.030
product_category_Coffee beans	18.1066	4.005	4.521	0.000	10.200	26.013
product_category_Drinking Chocolate	4.0021	1.675	2.389	0.018	0.695	7.309
product_category_Flavours	0.6722	1.548	0.434	0.665	-2.384	3.729
product_category_Loose Tea	0.3201	4.566	0.070	0.944	-8.694	9.334
product_category_Packaged Chocolate	-30.6559	8.565	-3.579	0.000	-47.564	-13.748
product_category_Tea	2.7452	0.780	3.519	0.001	1.205	4.285

```
=====
Omnibus:            21.756      Durbin-Watson:      1.820
Prob(Omnibus):      0.000      Jarque-Bera (JB):   42.217
Skew:               0.576      Prob(JB):           6.80e-10
Kurtosis:           5.066      Cond. No.           2.33e+03
=====
```

These predictor variables (independent) contribute to 98.3% of daily revenue!!

Results

- **Coffee products drive the strongest revenue**

Coffee beans show the highest positive impact on daily sales, making them the most influential product category.

- **Packaged chocolate reduces revenue**

Each additional unit sold is associated with lower daily revenue, suggesting a cannibalization effect with higher-margin items.

- **Evening hours generate the most revenue**

Evening time shows the strongest sales performance, followed by morning; afternoons consistently generate the least.

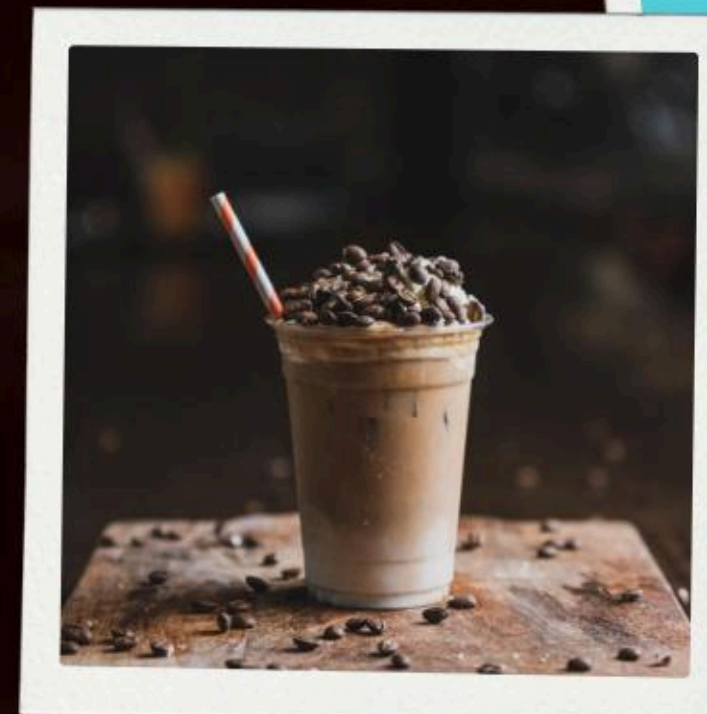
- **Findings guide smarter product and timing decisions**

These insights support prioritizing high-impact categories, adjusting merchandising, and designing time-specific promotions.



Insights

- 01 Revenue doubled:**
Daily revenue grew from \$2,500 to \$5,000 throughout 2023, demonstrating strong business growth
- 02 Balanced location performance:**
All three NYC locations perform similarly, with Hell's Kitchen leading at \$236K total revenue
- 03 Morning peak dominates:**
Morning Time generates highest revenue (\$600+ daily); evening drops significantly to \$236
- 04 Monday-Tuesday pattern:**
Monday shows strongest sales; Tuesday presents opportunity for targeted promotions to boost slower-day revenue



Recommendations



- 01 Optimize morning operations:**
Prioritize staffing and inventory during peak hours (6 AM-12 PM) for maximum efficiency
- 02 Target Tuesday promotions:**
Launch special offers on slowest-performing day to balance weekly revenue
- 03 Loyalty program focus:**
Introduce morning commuter loyalty program to increase repeat business and customer retention
- 04 Extend breakfast menu:**
Add popular breakfast items to afternoon hours to capture additional revenue opportunities
- 05 Dynamic pricing strategy:**
Implement off-peak pricing during afternoon/evening hours to drive traffic during slower periods

Managerial implications

- 01 Resource allocation:** Shift staff and supplies to morning shifts when customer demand peaks, improving service efficiency
- 02 Marketing strategy:** Design day-specific promotions (especially Tuesday) to balance weekly revenue and reduce sales volatility
- 03 Menu optimization:** Focus on quick-service breakfast items during peak hours to reduce wait times and maximize throughput
- 04 Cost management:** Minimize staffing during low-traffic afternoon/evening hours while maintaining quality service standards
- 05 Expansion planning:** Consistent performance across three locations validates business model replicability for strategic growth into new neighborhoods



Problems and limitations

01 Statistical Issues:

- Overfitting risk may reduce model accuracy on future data
- Coefficients for days of the week are statistically insignificant.

03 External Factors:

- Analysis doesn't account for weather, holidays, events, or competitor activities

04 Geographic Scope:

- Limited to three NYC locations; findings may not apply to other markets

02 Data Constraints:

- Dataset covers only 2023, missing long-term trends and seasonal patterns
- No customer demographic data available for targeted marketing strategies



Thank you

